

Homework 2

● Graded

Student

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Total Points

31 / 31 pts

Question 1

Problem 1

3 / 3 pts

✓ - 0 pts Correct

- 1 pt wrong number of ways to choose m red balls out of r red balls
- 1 pt wrong number of ways to choose k-m other colored balls out of b-r total other colored balls
- 1 pt wrong number of ways to choose k balls out of b balls
- 0.5 pts notation error: should be k-m, not m-k
- 1 pt should not be permutation
- 1 pt you should use notations given in the problem
- 3 pts cannot find answer to problem 1
- 1 pt not picking 1

Question 2

Problem 2

7 / 7 pts

✓ - 0 pts Correct

- 7 pts cannot find answer to problem 2
- 1 pt wrong total ways to choose 6 people from 16 people
- 1 pt (a) wrong ways to choose 3 women from 8 women
- 1 pt (a) wrong ways to choose 3 men out of 8 men
- 1 pt (b) wrong ways to choose 0 women and 6 men out of 8 men
- 1 pt (b) forget to take compliment of probability of choosing 0 women
- 1 pt (c) wrong ways to choose 4 people of out the 14 people excluding Emma and Jack
- 1 pt forget to take probability
- 2 pts (b) the question is not asking for the probability of choosing 1 woman and 5 men
- 2 pts (c) not correct
- 1 pt you should use numbers from the question
- 1 pt (b) wrong compliment
- 1 pt (c) should be 2 choose 2 for choosing Emma and Jack
- 1 pt not permutation
- 1 pt (c) there should not be 2!
- 1 pt (a) mutually exclusive events should be multiplied, not added together
- 1 pt (c) choosing emma and jack should be 2 choose 2

Question 3

Problem 3

6 / 6 pts

✓ - 0 pts Correct

- 1 pt Numerator for part b should be $3!3!2!1!$
- 1 pt Numerator for part c should be $4!3!$
- 2 pts incorrect answer and no work shown
- 2 pts incorrect methodology for part b
- 2 pts answers are not in probability form (a number instead of being between 0 and 1)
- 1 pt incorrect denominator
- 6 pts no answer submitted

Question 4

Problem 4

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts (a) wrong P(B)
- 0.5 pts (a) wrong $P(A \wedge B)$
- 0.5 pts (a) wrong answer
- 0.5 pts (b) wrong numerator
- 0.5 pts (b) wrong denominator
- 0.5 pts (c) wrong numerator
- 0.5 pts (c) wrong denominator
- 5 pts Incorrect
- 3.5 pts Show work
- 1.5 pts (a) Show work

Question 5

Problem 5

5 / 5 pts

✓ - 0 pts Correct

- 2 pts Answer given but did not show how to solve series
- 2 pts incorrect simplification
- 3 pts did not solve series
- 0.5 pts minor algebraic error (solved series correctly but forgot to multiply by 2)
- 0.5 pts could simplify further
- 2 pts incorrect setup. Should be: $P(Tail) = \sum_{n=1}^{\infty} P(Tail|coin\ n)P(coin\ n)$
- 5 pts problem not submitted

✓ - 0 pts Correct

- 0.5 pts Goal: Find $P(\text{non-spam} | \text{detected spam})$

- 0.5 pts formula of $P(\text{non-spam} | \text{detected spam})$

- 0.5 pts $P(\text{detected spam} | \text{spam}) = 0.99$

- 0.5 pts $P(\text{detected spam} | \text{non-spam}) = 0.05$

- 0.5 pts Expression for numerator : $P(\text{non-spam} \& \text{detected spam})$

- 0.5 pts Calculation of $P(\text{non-spam} \& \text{detected spam})$

- 0.5 pts Expression for denominator : $P(\text{detected spam})$

- 0.5 pts Formula of $P(\text{detected spam})$

- 0.5 pts Calculation of $P(\text{detected spam})$

- 5 pts Incorrect

1.
$$\frac{\binom{r}{m} \binom{b-r}{k-m}}{\binom{b}{k}}$$

$\binom{b}{k} \rightarrow$ ways to choose k balls from b

$\binom{r}{m} \rightarrow$ ways to choose m balls from r

$\binom{b-r}{k-m} \rightarrow$ ways to choose $k-m$ balls (non-red)
from $b-r$ balls (non-red)

✱

2. a)

$$\frac{\binom{8}{3} \binom{8}{3}}{\binom{16}{6}}$$

$$\binom{16}{6} = \frac{16!}{6! \cdot 10!}$$

8 women

8 men

= 16 total

$$b) \frac{\binom{8}{6}}{\binom{16}{6}} = P(\text{all men})$$

↗ No women

$$P(\text{at least 1 woman}) = 1 -$$

$$\frac{\binom{8}{6}}{\binom{16}{6}}$$

✱

c)

$$\frac{\binom{14}{4}}{\binom{16}{6}}$$

- Probability of choosing at least one woman, calculate the complement, which is having no women.

$$P(\text{at least 1 woman}) = 1 - P(\text{all men})$$

*

$$3. \quad b) \quad \frac{3! \cdot 3! \cdot 2!}{6!} = \frac{72}{720} = \boxed{\frac{1}{10}}$$

$$c) \quad \frac{4! \cdot 3!}{6!} = \frac{144}{720} = \boxed{\frac{1}{5}}$$

6! → Total arrangements

3 novels

2 math

1 Chem. book

3 things to arrange

↳ 2! math ↳ 3! novels
 ↳ 3! Chem. and other 2 categories

↳ # of ways to arrange 3 things

b) [1 Chem book - 1 unit/block
 2 Math books - 1 unit/block
 3 Novels - 1 unit/block] → 3!

c) [1 Chem book - 1 unit
 1st math book - 1 unit
 2nd math book - 1 unit
 3 novels - 1 unit] → 4!

↳ Within math book → 3! 2! 3!
 Unit is 2!
 ↳ Within novel unit is 3!
 ↳ within novel unit

$$P(A) = P(A \cap B) + P(A \cap B')$$

$$\hookrightarrow P(A \cap B) = P(A) - P(A \cap B')$$

$$\hookrightarrow P(A \cap B) = 0.4 - 0.1 = 0.3$$

$$P(B) = 1 - 0.3 = 0.7$$

$$P(A') = 1 - 0.4 = 0.6$$

Hw2 con.

$$4. \quad P(A) = 0.4, \quad P(B') = 0.3, \quad P(A \cap B') = 0.1$$

$$a) \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.4 + 0.7 - 0.3 = \boxed{0.8}$$

$$b) \quad P(B' | A) = \frac{P(A \cap B')}{P(A)} = \frac{0.1}{0.4} = \boxed{0.25} \quad \nearrow 1/4$$

$$c) \quad P(B | A') = \frac{P(A' \cap B)}{P(A')} = \frac{0.4}{0.6}$$

$$= \frac{2}{3} = \boxed{0.67}$$

$$P(B) = P(A \cap B) + P(A' \cap B)$$

$$\hookrightarrow P(A' \cap B) = P(B) - P(A \cap B)$$

$$\hookrightarrow 0.7 - 0.3 = 0.4$$

$$5. P(n) = \frac{2}{3^n}$$

$$P(\text{Tail} | n) = \frac{1}{2^n}$$

$$P(\text{Tail}) = \sum_{n=1}^{\infty} P(n) \cdot P(\text{Tail} | n) \rightarrow P(\text{Tail}) = \sum_{n=1}^{\infty} \frac{2}{3^n} \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} \frac{2}{6^n}$$

$$P(\text{Tail}) = 2 \sum_{n=1}^{\infty} \frac{1}{6^n} = \frac{1}{6} = \frac{1}{6} = \frac{1}{6} \cdot \frac{6}{5} = \frac{1}{5}$$

$$\frac{2}{1} \cdot \frac{1}{5} = \frac{2}{5} \checkmark$$

The Probability that the
result is Tail is: $\boxed{\frac{2}{5}}$

50/50
Chance

$$6. P(\text{Spam}) = 0.5 \rightarrow 50, P(\text{Non-spam}) = 1 - P(\text{Spam}) = 1 - 0.5 = 0.5$$

$$P(\text{Detected spam} | \text{spam}) = 0.99$$

$$\frac{0.05 \cdot 0.5}{0.52} = \frac{0.025}{0.52} = \boxed{0.0481}$$

$$P(\text{Detected spam} | \text{Non-spam}) = 0.05$$

$$P(\text{Non-spam} | \text{Detected spam}) = \frac{P(\text{Detected spam} | \text{Non-spam}) \cdot P(\text{Non-spam})}{P(\text{Detected spam})}$$

$$P(\text{Detected spam}) = P(\text{Detected spam} | \text{spam}) \cdot P(\text{spam}) + P(\text{Detected spam} | \text{Non-spam}) \cdot P(\text{Non-spam})$$

$$\hookrightarrow P(\text{Detected spam}) = (0.99 \cdot 0.5) + (0.05 \cdot 0.5) = 0.495 + 0.025 = 0.52$$

